

Lab Assessment Rubrics of Digital System Design (CPE-344)										
Lab Number			04							
Lab Title			Designing of Multiplexers using different modeling styles and Implement it on Spartan 6 FPGA board							
Submitted by										
Names			Registration Number							
Sumaira Safeer			FA22-BCE-019							
Rubrics name & number							Marks			
							In-Lab	Post-Lab		
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and then using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

Lab: 04

Objectives:

The objective of this lab is to get familiar with implementation of different kinds of multiplexer using

- Conditional Operator
- Reduction Operator
- Function Calling Method

Lab Tasks:

Task#01

- **Truth Table**

The truth table of is given below:

a	b	sel	Out
1	0	1	a
0	1	0	b

- **Verilog Code**

```
module mux2x1ConditionalOperator(out, a, b, sel);
input a, b, sel;
output out;
assign out = sel?a:b;
endmodule
```

- **Verilog Test Bench**

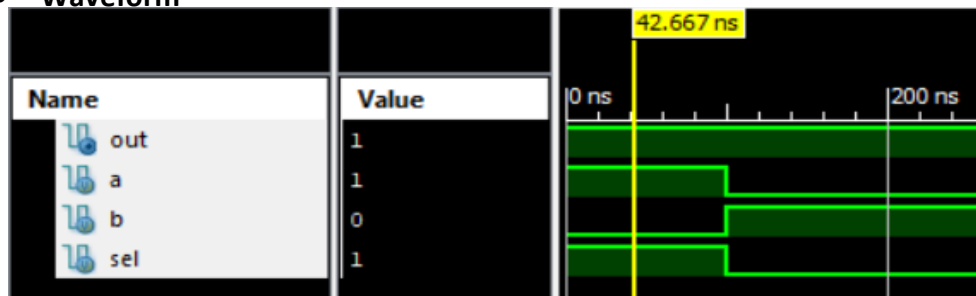
```
module mux2x1COTestBranch;
    reg a;
    reg b;
    reg sel;
    wire out;
    mux2x1ConditionalOperator uut (
        .out(out),
        .a(a),
        .b(b),
        .sel(sel)
    );
    initial begin
        a = 1;
        b = 0;
        sel = 1;
        #100;
        a = 0;
        b = 1;
    end
endmodule
```

```

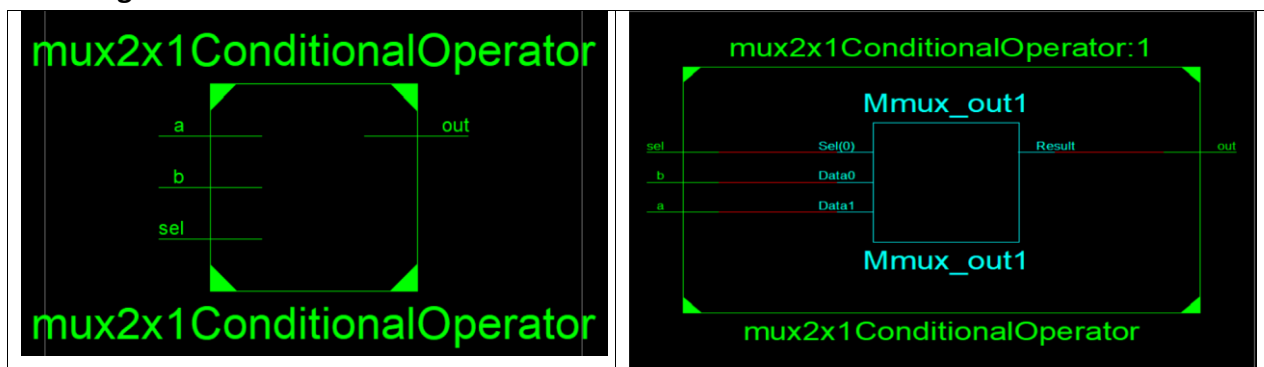
        sel = 0;
    end
endmodule

```

- Waveform



- RTL Diagram



Task#02

- Verilog Code

```

module mux4x1CO(out, a, b, c, d, sel);
    input a, b, c, d;
    input [1:0]sel;
    output out;
    assign out = sel[0]?(sel[1]?d:b):(sel[1]?c:a);
endmodule

```

- Verilog Test Bench

```

module mux4x1COTestBench;
    reg a;
    reg b;
    reg c;
    reg d;
    reg [1:0] sel;
    wire out;
    mux4x1CO uut (
        .out(out),
        .a(a),
        .b(b),

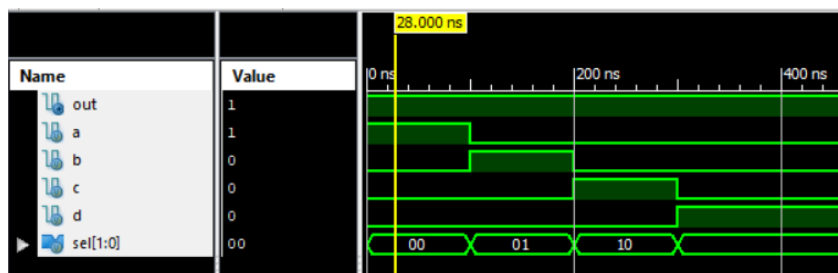
```

```

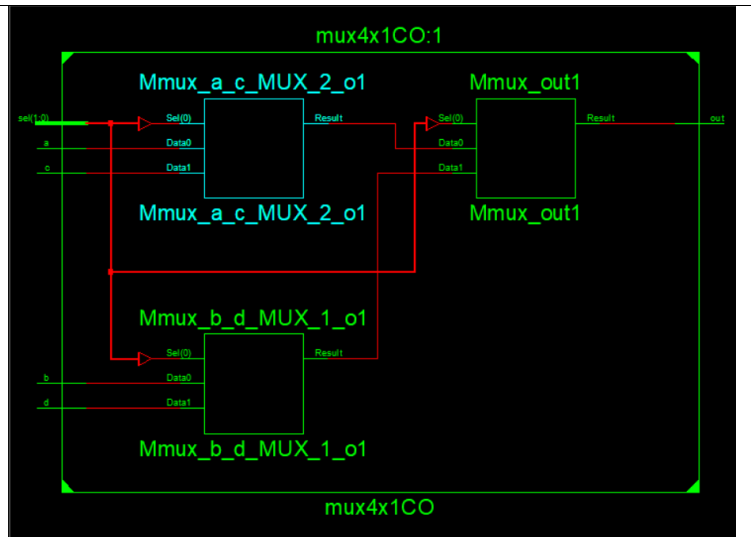
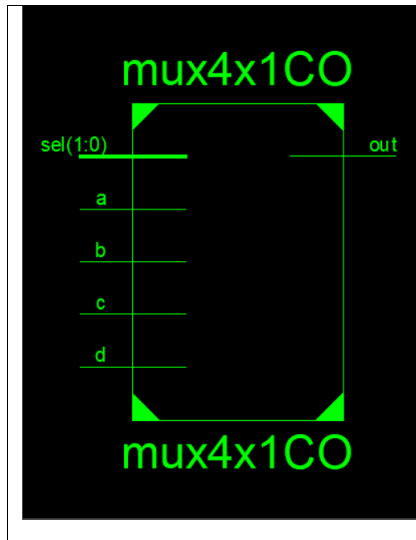
        .c(c),
        .d(d),
        .sel(sel)
    );
    initial begin
        a = 1;
        b = 0;
        c = 0;
        d = 0;
        sel = 2'b00;
        #100;
        a = 0;
        b = 1;
        c = 0;
        d = 0;
        sel = 2'b01;
        #100;
        a = 0;
        b = 0;
        c = 1;
        d = 0;
        sel = 2'b10;
        #100;
        a = 0;
        b = 0;
        c = 0;
        d = 1;
        sel = 2'b11;
    end
endmodule

```

- Waveform



- RTL Diagram



Task#02

- Verilog Code

```
module mux8x1OS(out, a, b, c, d, e, f, g, h, sel);
input a, b, c, d, e, f, g, h;
input [2:0]sel;
output out;
assign out = sel[2]?(sel[1]?(sel[0]?h:g):(sel[0]?f:e)):(sel[1]?(sel[0]?d:c):(sel[0]?b:a));
endmodule
```

- Verilog Test Bench

```
module mux8x1TestBench;
reg a, b, c, d, e, f;
reg g, h;
reg [2:0] sel;
wire out;
mux8x1OS uut (
.out(out),
.a(a),.b(b),
.c(c), .d(d),
.e(e), .f(f),
.g(g), .h(h),
.sel(sel)
);
initial begin
a = 1; b = 0;
c = 0; d = 0;
e = 0; f = 0;
g = 0; h = 0;
sel = 3'b000;
#100;
a = 0; b = 1;
c = 0; d = 0;
e = 0; f = 0;
```

```
g = 0; h = 0;  
sel = 3'b001;  
#100;
```

```
a = 0; b = 0;  
c = 1; d = 0;  
e = 0; f = 0;  
g = 0; h = 0;  
sel = 3'b010;  
#100;
```

```
a = 0; b = 0;  
c = 0; d = 1;  
e = 0; f = 0;  
g = 0; h = 0;  
sel = 3'b011;  
#100;
```

```
a = 0; b = 0;  
c = 0; d = 0;  
e = 1; f = 0;  
g = 0; h = 0;  
sel = 3'b100;  
#100;
```

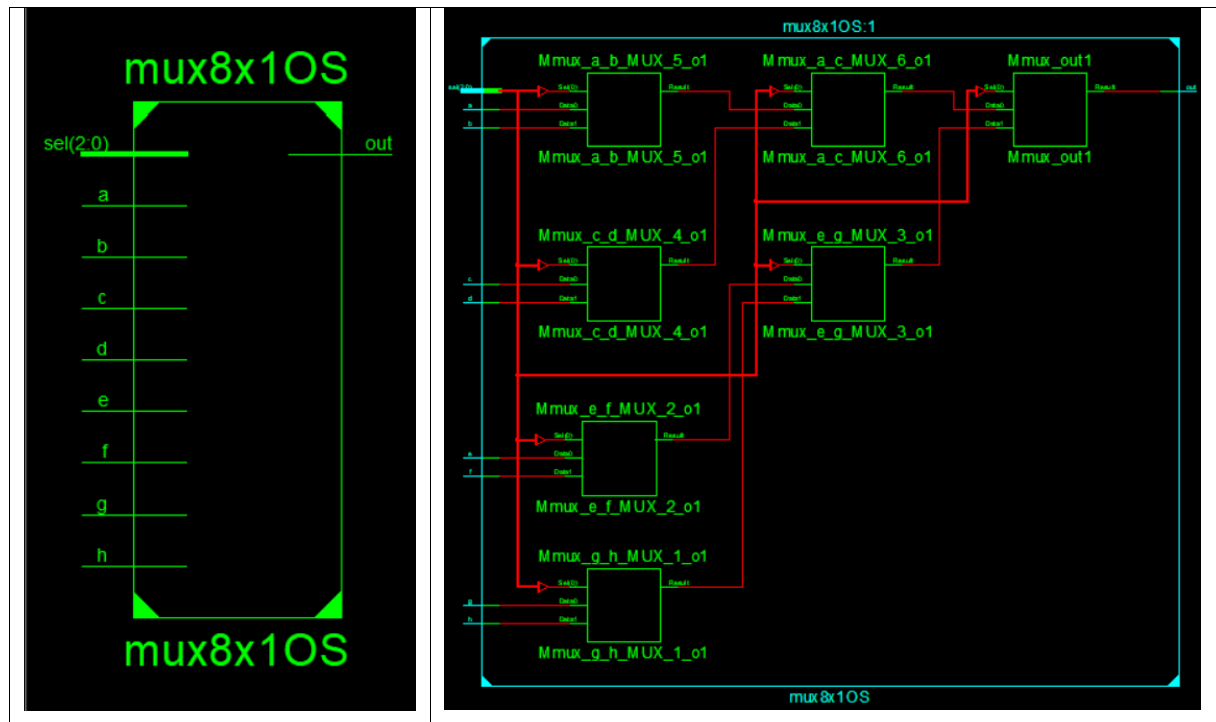
```
a = 0; b = 0;  
c = 0; d = 0;  
e = 0; f = 1;  
g = 0; h = 0;  
sel = 3'b101;  
#100;
```

```
a = 0; b = 0;  
c = 0; d = 0;  
e = 0; f = 0;  
g = 1; h = 0;  
sel = 3'b110;  
#100;
```

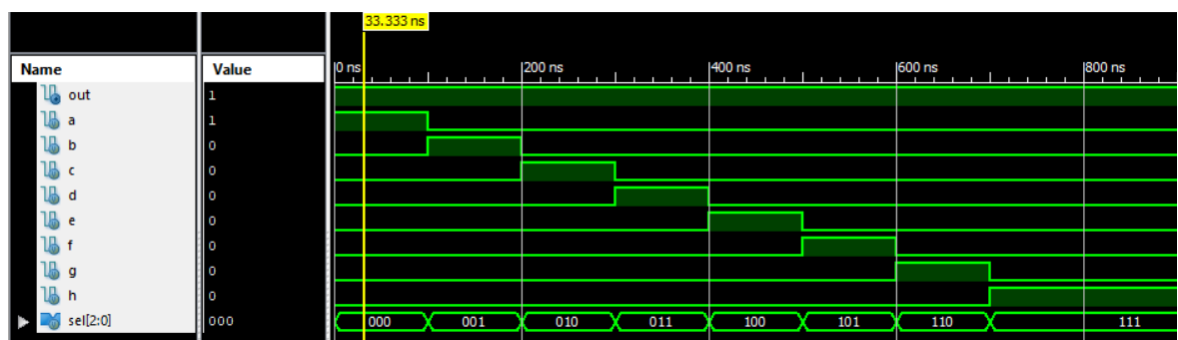
```
a = 0; b = 0;  
c = 0;  
d = 0; e = 0;  
f = 0; g = 0;  
h = 1;  
sel = 3'b111;  
#100;
```

```
end  
endmodule
```

- RTL Systematic



- Waveform



Task#04

- Verilog Code

```
module mux2x1ReductionOperator(out, a, b, sel);
input a, b, sel;
output out;
assign out= (a&~sel)|(b&sel);
endmodule
```

- Verilog Test Bench

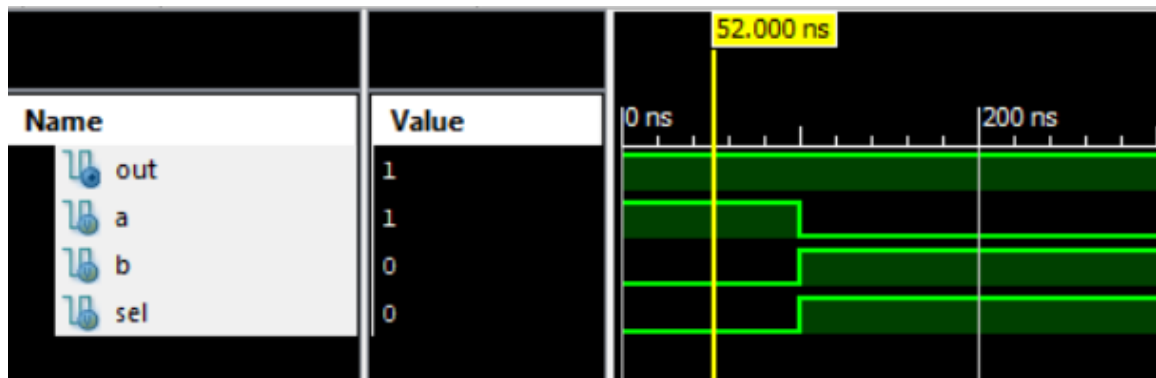
```
module mux2x1ReductionOperatorTestBench;
reg a;
reg b;
reg sel;
wire out;
```

```

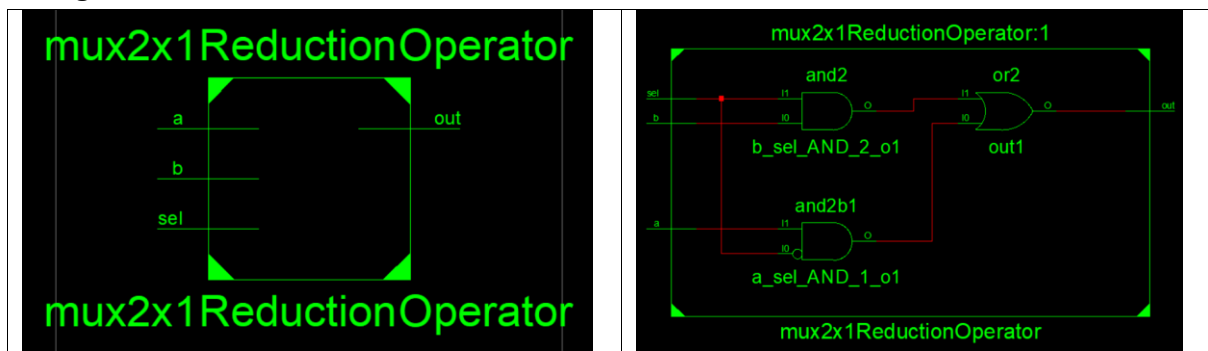
mux2x1ReductionOperator uut (
    .out(out),
    .a(a), .b(b), .sel(sel)
);
initial begin
    a = 1; b = 0; sel = 0;
    #100;
    a = 0; b = 1; sel = 1;
    #100;
end
endmodule

```

Waveform



• RTL Diagram



Task#05

• Verilog Code

```

module mux4x1ReductionOperator(out, a, b, c, d, sel);
    input a, b, c, d;
    input [1:0]sel;
    output out;
    assign out = (a&~sel[0]&~sel[1])|(b&~sel[1]&sel[0])|(c&sel[1]&~sel[0])|(d&sel[1]&sel[0]);
endmodule

```

• Verilog Test Bench

```

module mux4x1ReductionOperatorTestBench;
    reg a, b, c, d;
    reg [1:0] sel;
    wire out;

```

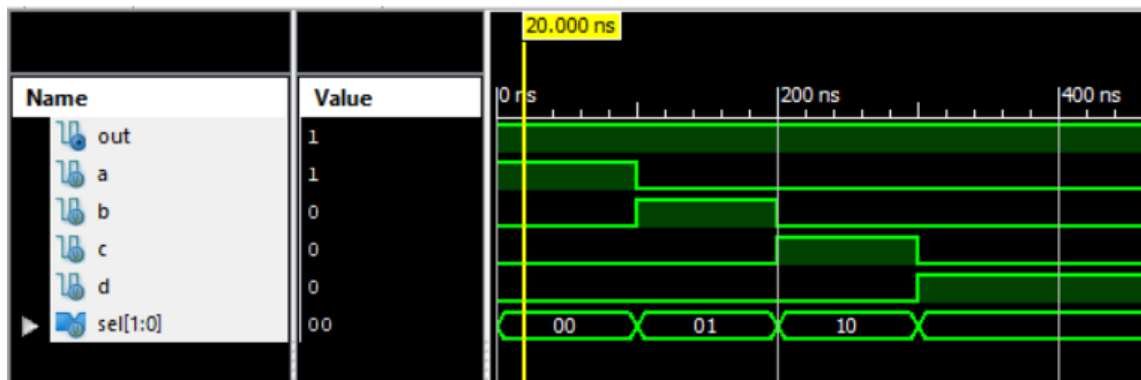


```

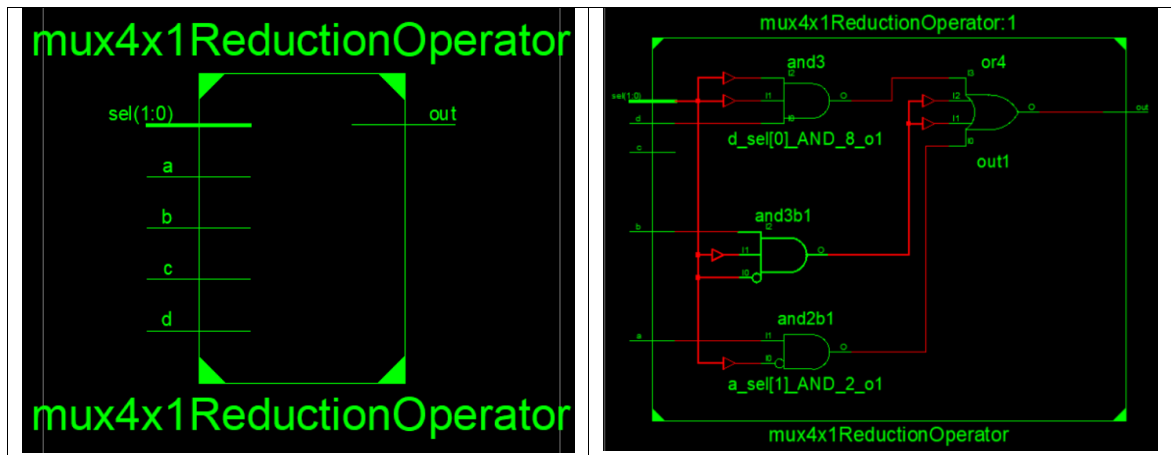
mux4x1ReductionOperator uut (
    .out(out),
    .a(a), .b(b),
    .c(c), .d(d),
    .sel(sel)
);
initial begin
    a = 1; b = 0;
    c = 0; d = 0;
    sel = 2'b00;
    #100;
    a = 0; b = 1;
    c = 0; d = 0;
    sel = 2'b01;
    #100;
    a = 0; b = 0;
    c = 1; d = 0;
    sel = 2'b10;
    #100;
    a = 0; b = 0;
    c = 0; d = 1;
    sel = 2'b11;
    #100;
end
endmodule

```

- Waveform



- RTL Diagram



Task#06

• Verilog Code

```
module mux8x1ReductionOperator(out, a, b, c, d, e, f, g, h, sel);
input a, b, c, d, e, f, g, h;
input [2:0]sel;
output out;

assign out = ((a&~sel[2])&~sel[1]&~sel[0])|
              ((b&~sel[2])&~sel[1]&sel[0])|
              ((c&~sel[2])&sel[1]&~sel[0])|
              ((d&~sel[2])&sel[1]&sel[0])|
              ((e&sel[2])&~sel[1]&~sel[0])|
              ((f&sel[2])&~sel[1]&sel[0])|
              ((g&sel[2])&sel[1]&~sel[0])|
              ((h&sel[2])&sel[1]&sel[0]);

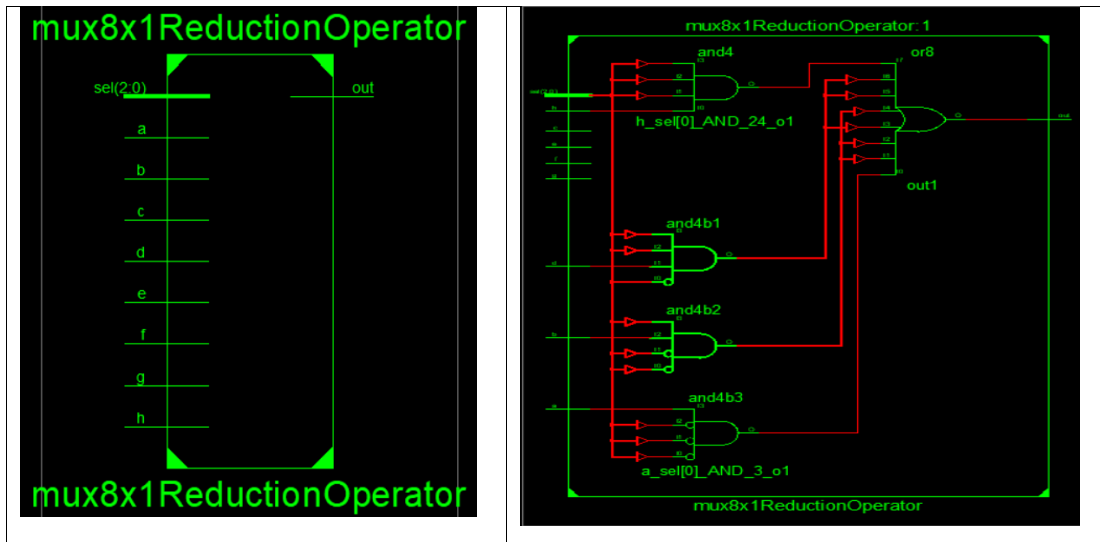
endmodule
```

• Verilog Test Bench

```
module mux8x1ReductionOperatorTestBench;
reg a,b,c,d,e,f,g,h;
reg [2:0] sel;
wire out;
mux8x1ReductionOperator uut (
    .out(out),
    .a(a), .b(b),
    .c(c), .d(d),
    .e(e), .f(f),
    .g(g), .h(h),
    .sel(sel)
);
initial begin
    a = 1; b = 0;
    c = 0; d = 0;
    e = 0; f = 0;
    g = 0; h = 0;
end
```

```
sel = 3'b000;
#100;
a = 0; b = 1;
c = 0; d = 0;
e = 0; f = 0;
g = 0; h = 0;
sel = 3'b001;
#100;
a = 0; b = 0;
c = 1; d = 0;
e = 0; f = 0;
g = 0; h = 0;
sel = 3'b010;
#100;
a = 0; b = 0;
c = 0; d = 1;
e = 0; f = 0;
g = 0; h = 0;
sel = 3'b011;
#100;
a = 0; b = 0;
c = 0; d = 0;
e = 1; f = 0;
g = 0; h = 0;
sel = 3'b100;
#100;
a = 0; b = 0;
c = 0; d = 0;
e = 0; f = 1;
g = 0; h = 0;
sel = 3'b101;
#100;
a = 0; b = 0;
c = 0; d = 0;
e = 0; f = 0;
g = 1; h = 0;
sel = 3'b110;
#100;
a = 0; b = 0;
c = 0; d = 0;
e = 0; f = 0;
g = 0; h = 1;
sel = 3'b111;
#100;
end
endmodule
```

- RTL Diagram



- Waveform

